FYS5429/9429 March 19

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$$X \longrightarrow \boxed{\text{Encode}} \;\Big|\; h \;\Big|\; \boxed{\text{Decoder}} \longrightarrow \hat{X}$$

latent space
(effective space)

AE : optimize the Reconstruction error

$$J = \frac{1}{N} \sum_{i=0}^{N-1} (x_i - \hat{x}_i)^2$$

$$= \frac{1}{N} \underbrace{\| x - \hat{x} \|_F^2}_{J} = \sum_{i,j} J_{ij}^2$$

$$= \frac{1}{N} \mathrm{Tr}\left[(x - \hat{x})^T (x - \hat{x}) \right]$$

Autoencoder (linear)

$$h = W_e X$$

$$\hat{X} = W_d h = W_d W_e X$$

$$\hat{W}_d, \hat{W}_e = \underset{W_d\, W_e}{\arg\min} \; \frac{1}{N} \| (X - W_d W_e X) \|_F^2$$

PCA :

Basic element : covariance matrix

$$X \in \mathbb{R}^{N \times P}$$

$$X = \left[\vec{x}_0 \ \vec{x}_1 \ \cdots \ \vec{x}_{p-1} \right]$$

$$\mathrm{cov}[\vec{x}_i, \vec{x}_j] = \frac{1}{N} \sum_{k=0}^{N-1} (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$\mathrm{cov}[X] = \Sigma = \frac{1}{N} X^T X$$

$$\mathrm{cov}[y] = S \Sigma S^T = D$$

$$S S^T = S^T S = \mathbb{1}$$

$$D = \begin{bmatrix} \lambda_0 & & & \\ & \lambda_1 & & \\ & & \ddots & \\ & & & \lambda_{p-1} \end{bmatrix}$$

$$\lambda_i = \sigma_i^2 \geq 0$$

$$\lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$$

$$k \leq p$$

<u>PCA</u>

$$\sum_{i=0}^{k-1} \frac{\lambda_i}{\sigma_i^2} \leq \text{cutoff} \leq \text{Total variance}$$

SVD

$$X \in \mathbb{R}^{N \times P}$$

$$X = U D' V^T$$

$$U \in \mathbb{R}^{N \times N}$$

$$D' \in \mathbb{R}^{N \times P}$$

$$V \in \mathbb{R}^{P \times P}$$

$$U U^T = U^T U = \mathbb{1}_{N \times N}$$

$$V V^T = V^T U = \mathbb{1}_{P \times P}$$

$$D' = \begin{bmatrix} \lambda'_0 & & & 0 \\ & \lambda_1 & & \\ 0 & & \ddots & \\ & & & \lambda_{p-1} \\ & & & & 0 \end{bmatrix}$$

$$D' = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} \vec{u}_0 & \vec{u}_1, & \cdots & \vec{u}_{N-1} \end{bmatrix}$$

$$\vec{u}_i^T \vec{u}_j = \delta_{ij} \quad \text{(same for } V\text{)}$$

SVD

$$\hat{X} = \left(\sum_{j=0}^{p-1} \vec{u}_j u_j^T \right) X$$

$$\simeq \left(\sum_{j=0}^{k-1} \vec{u}_j u_j^T \right) X$$

$$k < p$$

The singular values squared are the eigen values of $\mathrm{cov}[X] = \Sigma$

$$\frac{1}{N} X^T X = \Sigma$$

eigenvectors $\vec{v}_i'$

eigenvalues $\lambda_i'^2$

$$\left(\frac{1}{N} X^T X\right) \vec{v}_i = \frac{\sigma_i^2}{\frac{\lambda_i'^2}{N}} \vec{v}_i'$$

Keep those $\vec{v}_i$ which correspond to largest σ_i^2

$$\min_{W_d W_e} \frac{1}{N} \| (X - \underbrace{W_d W_e}_{S S^T} X) \|_F^2$$

PCA or SVD

$$S = [\vec{s}_0, \vec{s}_1, \ldots \vec{s}_k \ldots \vec{s}_{p-1}]$$

$$S S^T = S^T S = \mathbb{1}_{p \times p}$$

$$(U U^T = \mathbb{1}_{N \times N} \quad V V^T = \mathbb{1}_{p \times p})$$

$$S = [\vec{s}_0 \ \vec{s}_1 \ \dots \ \vec{s}_k \ \dots \ \vec{s}_{p-1}]$$

$$\vec{s}_0^T \vec{s}_0 = 1 \qquad \vec{s}_0 \vec{s}_0^T$$

$$\hat{X} = \vec{s}_0 \vec{s}_0^T X \qquad (\text{WaWeX})$$

$$\| (X - s_0 s_0^T X) \|_2^2 \leq \varepsilon$$

we define an optimization problem of the type

$$s_0^T \underbrace{\mathrm{Cov}[X]}_{\Sigma} s_0 + \lambda_0 (1 - s_0^T s_0)$$

optimize wrt s_0^T

$$\mathrm{cov}[x]\, s_0 = \Sigma s_0 = \lambda_0 s_0$$

multiply from left with s_0^T

$$s_0^T \mathrm{cov}[x]\, s_0 = \lambda_0$$

$$\left(\text{SVD} : \quad \Sigma \vec{v}_i = \sigma_i^2 \vec{v}_i \right)$$

variance is $\frac{\lambda_0}{N} = \sigma_0^2$

called the first principal component (PCA)

Next s_1 by induction

$$s_1^T s_0 = 0 \qquad s_1^T s_1 = 1$$

$$\mathcal{L} = s_1^T \Sigma s_1 + \lambda_1 (1 - s_1^T s_1) + \gamma s_1^T s_0$$

optimize wrt s_1^T, λ_1, γ

s_1^T : $\Sigma s_1 - \gamma s_0/2 = \lambda_1 s_1$

multiply from left with s_0^T

$$\underbrace{s_0^T \Sigma}_{\underbrace{s_0^T \lambda_0}_{=0}} s_1 + \gamma/2 \underbrace{s_0^T s_0}_{1} = \lambda_1 \underbrace{s_0^T s_1}_{=0}$$

$$\Rightarrow \gamma = 0$$

$$s_1^T \Sigma s_1 = \lambda_1 = \sigma_1^2$$

the second variance

$$\sigma_0^2 \geq \sigma_1^2$$

By induction

$$\mathcal{L} = s_i^T \Sigma s_i$$

$$+ \lambda_i (1 - s_i^T s_i) +$$

$$\sum_{j=0}^{i-1} \gamma_j s_i^T s_j$$

PCA :

$$\hat{X} = \left(\sum_{j=0}^{k-1} s_j s_j^T \right) X$$

AE - linear

$$A = W_d W_e = \left(\sum_{j=0}^{k-1} s_j s_j^T \right)$$

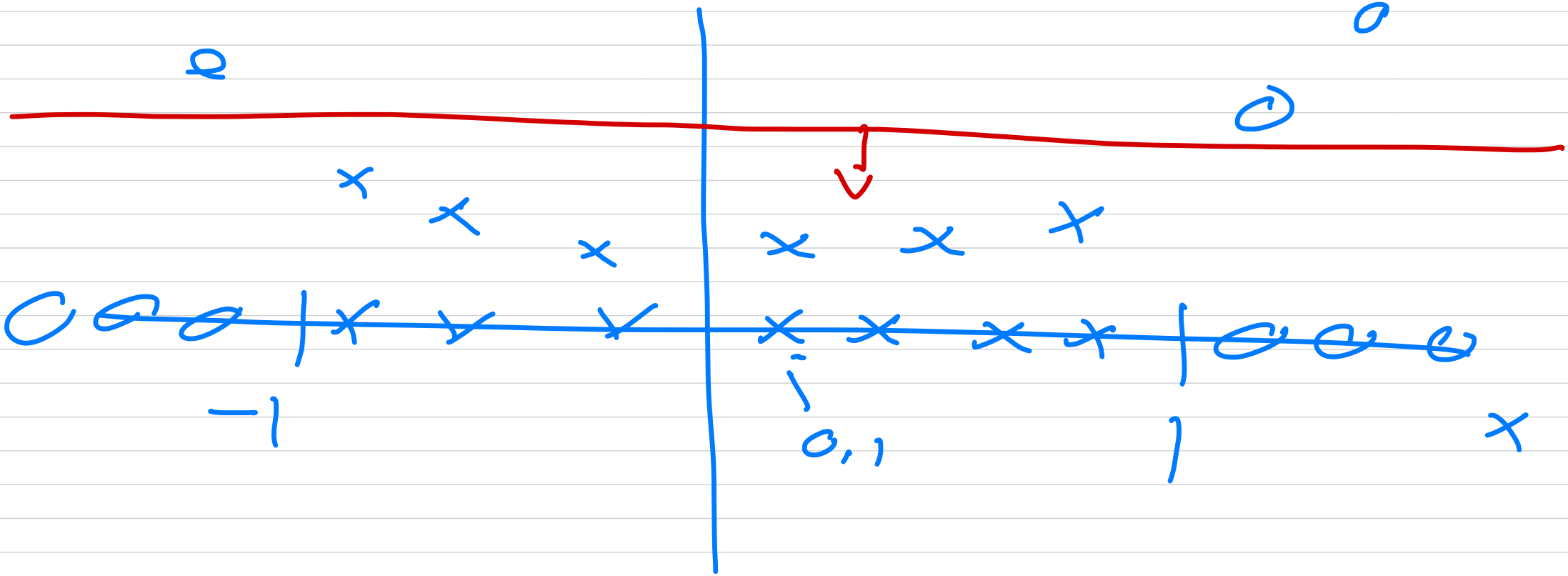
SVD

$$A = \left(\sum_{j=0}^{k-1} u_j u_j^T \right)$$

$k \leq p$

$$X = U D' V^T$$

Kernel transformations



Binary classification

simply $x \to x^2$ (kernel)

e^{x^2} ...